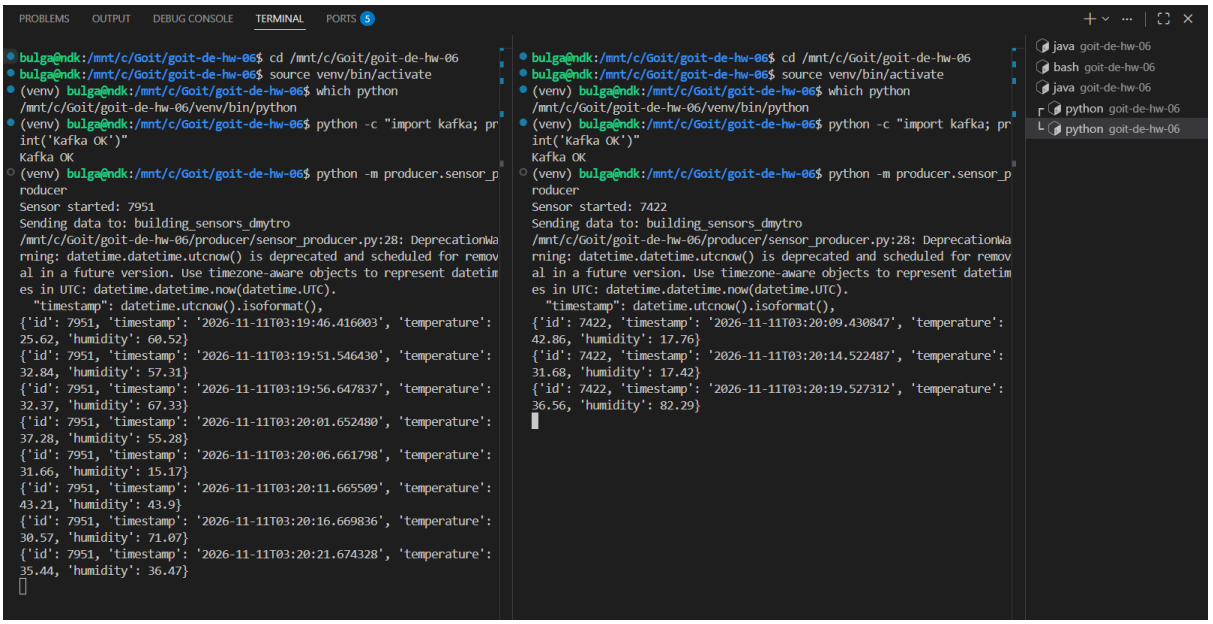
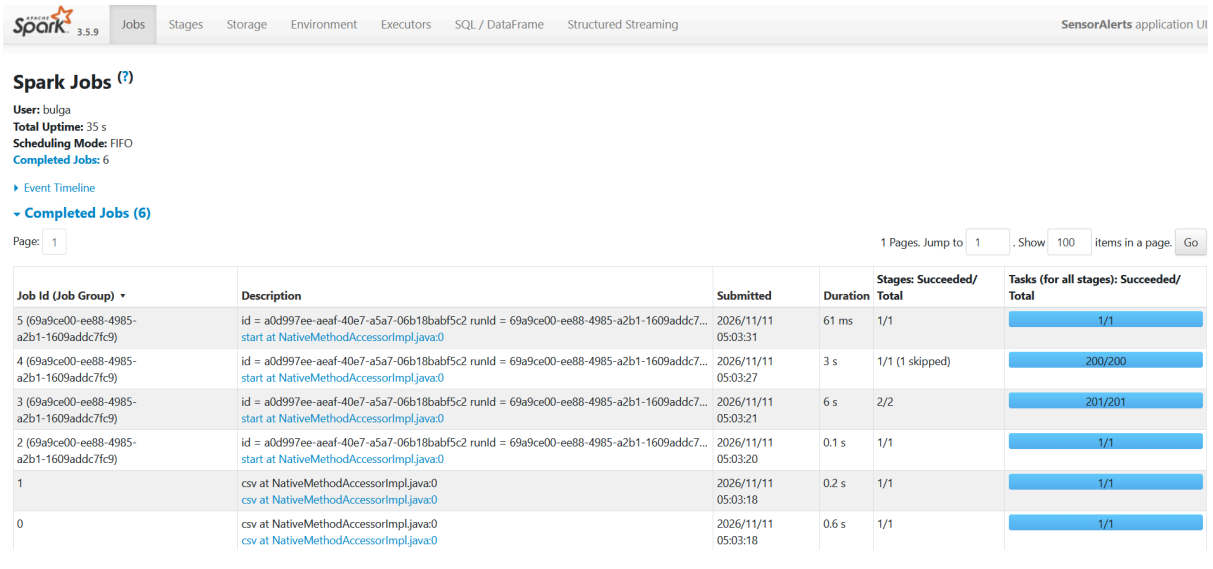
1. Працююча Kafka і Топіки:

```
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ python -c "import kafka; print(kafka.__version__)"
3.0.13
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ which python
/mnt/c/Goit/goit-de-hw-06/venv/bin/python
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ pip freeze > requirements.txt
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ python -c "from kafka import KafkaProducer; print('OK')"
OK
(venv) bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ python create_topics.py
Topics created
```

```
bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ jps
532842 Jps
520798 SparkSubmit
530884 ConsoleConsumer
377850 Kafka
bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ ~/kafka/bin/kafka-topics.sh --list --bootstrap-server localhost:9092
__consumer_offsets
alerts_dmytro
building_sensors_dmytro
bulga@ndk:/mnt/c/Goit/goit-de-hw-06$
```

2. Генерація даних сенсорів та відправка даних в `building_sensors` з демонстрацією двох (або більше) одночасних сеансів роботи двох запусків програми;





3. Демонстрація того, що відфільтровані дані були послані у відповідні топіки.

```
bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ spark-submit \
--packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.5.9 \
spark_alert_processor.py
(driver, 10.255.255.254, 43701, None)
26/11/11 05:03:08 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 10.255.255.254, 43701, None)
26/11/11 05:03:08 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 10.255.255.254, 43701, None)
26/11/11 05:03:19 WARN ResolveWriteToStream: Temporary checkpoint location created which is deleted normally when the query didn't f
ail: /tmp/temporary-d1a6390b-724f-4366-89c3-d24640ed26c8. If it's required to delete it under any circumstances, please set spark.sq
l.streaming.forceDeleteTempCheckpointLocation to true. Important to know deleting temp checkpoint folder is best effort.
26/11/11 05:03:19 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will b
e disabled.
26/11/11 05:03:19 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will b
e disabled.
26/11/11 05:03:19 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.
records, auto.offset.reset]' were supplied but are not used yet.
26/11/11 05:03:20 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.
records, auto.offset.reset]' were supplied but are not used yet.
-------------------------------------------
Batch: 0
-------------------------------------------
+------+-----+-----+----+-------+---------+
|window|t_avg|h_avg|code|message|timestamp|
+------+-----+-----+----+-------+---------+
+------+-----+-----+----+-------+---------+

-------------------------------------------
Batch: 1
-------------------------------------------
+------------------------------------------+------------------+-----------------+----+-----------+----------------------+
|window                                    |t_avg             |h_avg            |code|message    |timestamp             |
+------------------------------------------+------------------+-----------------+----+-----------+----------------------+
|{2026-11-06 03:48:30, 2026-11-06 03:49:30}|35.013999999999996|36.678           |101 |It's too dry|2026-11-11 05:03:27.619|
|{2026-11-06 04:34:30, 2026-11-06 04:35:30}|30.656666666666666|64.64666666666668|102 |It's too wet|2026-11-11 05:03:27.619|
+------------------------------------------+------------------+-----------------+----+-----------+----------------------+
```

```
bulga@ndk:/mnt/c/Goit/goit-de-hw-06$ ~/kafka/bin/kafka-console-consumer.sh \
  --bootstrap-server localhost:9092 \
  --topic alerts_dmytro \
  --from-beginning
  The consumer rebalance protocol (KIP-848) is production-ready! Set group.protocol=consumer to try it out. See https://kafka.apache.o
  rg/documentation/#consumer_rebalance_protocol
  {"window":{"start":"2026-11-06T03:48:30.000+02:00","end":"2026-11-06T03:49:30.000+02:00"},"t_avg":35.013999999999996,"h_avg":36.678,
  "code":101,"message":"It's too dry","timestamp":"2026-11-11T04:49:17.099+02:00"}
  {"window":{"start":"2026-11-06T04:34:30.000+02:00","end":"2026-11-06T04:35:30.000+02:00"},"t_avg":30.656666666666666,"h_avg":64.6466
  6666666668,"code":102,"message":"It's too wet","timestamp":"2026-11-11T04:49:17.099+02:00"}
```